

HDOC - Day 6: Complete Answers

Question 1

Write a C program to find the breadth, perimeter, and diagonal of a rectangle when its area and length are given.

1. Understand the problem

Input: area and length of the rectangle. Output: breadth, perimeter and diagonal.

2. Identify formulas

Breadth = Area / Length

Perimeter = 2 x (Length + Breadth)

Diagonal = $\sqrt{\text{Length}^2 + \text{Breadth}^2}$

3. Algorithm

Start.

Read area and length.

Calculate breadth = area / length.

Calculate perimeter = 2 x (length + breadth).

Calculate diagonal = $\sqrt{\text{length} \times \text{length} + \text{breadth} \times \text{breadth}}$.

Display breadth, perimeter and diagonal.

Stop.

4. C Program

```
#include <stdio.h>
#include <math.h>

int main() {
    float area, length, breadth, perimeter, diagonal;

    printf("Enter area and length: ");
    scanf("%f %f", &area, &length);

    breadth = area / length;
    perimeter = 2 * (length + breadth);
    diagonal = sqrt(length * length + breadth * breadth);

    printf("Breadth = %.2f\n", breadth);
    printf("Perimeter = %.2f\n", perimeter);
    printf("Diagonal = %.2f\n", diagonal);
    return 0;
}
```

5. Sample calculation

For area = 48 cm² and length = 8 cm: breadth = 6 cm, perimeter = 28 cm, diagonal = 10 cm.

6. Result

The program successfully calculates the breadth, perimeter and diagonal of the rectangle.

Question 2

Write a C program to find the coloring cost, area, and perimeter of a kite-shaped design when its diagonals, unequal adjacent sides, and coloring rate per square centimeter are given.

1. Understand the problem

Input: diagonals d1 and d2, unequal adjacent sides a and b, and coloring rate per square centimeter.

Output: area, perimeter and coloring cost.

2. Identify formulas

Area = $(d1 \times d2) / 2$

Perimeter = $2 \times (a + b)$

Coloring Cost = Area x Rate

3. Algorithm

Start.

Read d1, d2, side a, side b and coloring rate.

Calculate area = $(d1 \times d2) / 2$.

Calculate perimeter = $2 \times (a + b)$.

Calculate coloring cost = area x rate.

Display area, perimeter and coloring cost.

Stop.

4. C Program

```
#include <stdio.h>

int main() {
    float d1, d2, a, b, rate;
    float area, perimeter, cost;

    printf("Enter two diagonals: ");
    scanf("%f %f", &d1, &d2);
    printf("Enter two unequal adjacent sides: ");
    scanf("%f %f", &a, &b);
    printf("Enter coloring rate per sq. cm: ");
    scanf("%f", &rate);

    area = (d1 * d2) / 2;
    perimeter = 2 * (a + b);
    cost = area * rate;

    printf("Area = %.2f sq. cm\n", area);
    printf("Perimeter = %.2f cm\n", perimeter);
    printf("Coloring Cost = %.2f\n", cost);
    return 0;
}
```

5. Sample calculation

For d1 = 10 cm, d2 = 8 cm, a = 5 cm, b = 6 cm and rate = Rs. 2 per cm²: area = 40 cm², perimeter = 22 cm and coloring cost = Rs. 80.

6. Result

The program successfully calculates the area, perimeter and coloring cost of the kite-shaped design.